

Metro Ticket Generating System

ServiceNow Global Certification Project — Phase 1 & Agile Planning Documentation

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Project Type: Individual Work

Platform: ServiceNow

Module: Global Certification Micro-Projects

1. Executive Summary & Project Overview

The **Metro Ticket Generating System** is an enterprise-grade ServiceNow application built to automate and streamline urban transit ticketing workflows. Traditional manual ticketing involves long queues, administrative bottlenecks, and rigid pricing models. This project implements a fully digital Service Catalog item coupled with backend business logic to deliver seamless ticket generation, dynamic fare calculation, and instant QR code verification.

Core Mission: To replace manual station ticketing with an automated ServiceNow workflow that handles dynamic passenger classifications, multi-station distance pricing, and secure digital code generation.

2. Agile Scrum & Sprint Planning Framework

Following agile delivery standards, the project scope is structured into Epics, Sprints, and User Stories (USN). Story points are estimated using the Fibonacci scale (1, 2, 3, 5) to gauge task complexity and team velocity.

Sprint 1: Architecture, Data Modeling & Backend Configuration (Epic 1 & 2)

Task ID	User Story / Epic Description	Complexity / Story Points
USN1	Gather client needs & define Station Master Data tables	2
USN2	Load initial station nodes & zone configurations	1
USN3	Handle missing transit parameters and input validation rules	3
USN4	Create custom dictionary fields for dynamic fare mapping	3
USN5	Handle data consistency across source and destination matrices	3
Total Story Points in Sprint 1:		12

Sprint 2: UI/UX Customization, QR Integration & Workflow Automation (Epic 3 & 4)

Task ID	User Story / Epic Description	Complexity / Story Points
USN6	Design Service Catalog UI form with dynamic variable sets	2
USN7	Configure Client Scripts for real-time form behavior	2
USN8	Implement Server-side Script Includes for Fare Logic calculations	2
USN9	Integrate QR Code generation engine for digital tickets	4
USN10	Develop Admin Dashboard for operational metrics & live tracking	5
USN11	End-to-End Testing, Flow Logs validation & UAT deployment	5
Total Story Points in Sprint 2:		20

Velocity Calculation & Performance Metrics

Team velocity measures the amount of work completed per sprint, providing predictability for iterative delivery.

Total Story Points = 12 (Sprint 1) + 20 (Sprint 2) = 32

Number of Sprints = 2

Velocity = $\frac{\text{Total Story Points}}{\text{Number of Sprints}} = \frac{32}{2} = 16 \text{ Story Points/Sprint}$

3. Stakeholder Mapping & Functional Scope

Precise stakeholder identification ensures that all user personas interact smoothly with the ServiceNow instance, maintaining robust security segregation and optimized operational controls.

Stakeholder Persona	Role & Access Level	Key System Interactions
Passengers (End Users)	Self-Service Portal User	Access Service Catalog, select route, book tickets, view digital QR pass.
Station Managers	Fulfillment / Support Agent	Verify ticket validity, inspect scanned QR logs, handle passenger queries.
Metro Operations Team	Operations Analyst	Monitor commuter traffic volumes, review daily travel analytics.
IT Administrators	System Administrator (admin)	Manage database tables, configure fare matrices, monitor Flow Designer execution logs.

4. Technical Execution Roadmap & Visual Architecture

The development lifecycle follows a structured sequential pipeline: *ext{Catalog Form Setup}* *ightarrow* *ext{Fare Logic Script}* *ightarrow* *ext{QR Code Integration}* *ightarrow* *ext{Station Mapping}* *ightarrow* *ext{UI Pages}* *ightarrow* *ext{UAT & Deployment}*.

[Visual Artifact 1: Service Catalog UI & Dynamic Form Behavior]
Captured: Form field dependencies (Source/Destination validation), Passenger Type selectors, and Real-time UI responsiveness.

[Visual Artifact 2: QR Code Generation & Flow Designer Execution Path]
Captured: Server-side script execution logs, QR payload rendering, and User Data Storing Process tables.

5. Conclusion & Future Enhancements

The Metro Ticket Generating System successfully demonstrates advanced ServiceNow application development capabilities, adhering to rigorous design patterns and enterprise standards. Rigorous UAT testing with realistic urban transit datasets confirms accuracy in dynamic fare computation and rapid QR generation.

Future Scope: Expansion avenues include SMS & Email ticketing notifications, automated payment gateway integration, and a comprehensive travel analytics dashboard for executive decision-making.